



The impact of parental burnout on family cohesion: the mediating roles of parent-child relationships and parental differences

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Abstract

Parenting is a complex and challenging process that can be stressful, potentially leading to parental burnout. Based on family systems theory, this study explored the impact of parental burnout on family cohesion and the mediating roles of parent-child relationships and parental differences. Participants included fathers, mothers, and children from 1,913 families. After controlling for family socioeconomic status and children's gender, the results revealed the following: (1) Mothers' parental burnout negatively predicted family cohesion, while fathers' parental burnout did not have a significant predictive effect. (2) Father-child relationship mediated the association between fathers' parental burnout and family cohesion, while mother-child relationship mediated the association between mothers' parental burnout and family cohesion. (3) Mothers' parental burnout also influenced family cohesion through father-child relationship. (4) The impact of mothers' parental burnout was stronger than that of fathers. In conclusion, these findings highlight the mechanisms underlying the relationship between parental burnout and family cohesion, emphasizing the more substantial negative effect of mothers' parental burnout, thereby contributing to the understanding of complex family dynamics.

Keywords Parental burnout · Parent-child relationships · Family cohesion · Parental differences

Introduction

Parental burnout is a state of intense exhaustion associated with the parental role (Roskam et al., 2017). It occurs when there is a long-term mismatch between the parenting-related stress perceived by parents and the resources available within the parenting domain (Mikolajczak & Roskam, 2018). Parents experiencing parental burnout are at a higher risk of developing depressive symptoms, addictive behaviors, sleep disorders, as well as escape and suicidal ideations compared to parents without burnout (Mikolajczak et al., 2018, 2019; Wang et al., 2021). In addition,

the consequences of parental burnout may be transmitted among family members, affecting partners and children, and negatively impacting the entire family (Gillis & Roskam, 2019; Mikolajczak et al., 2018; Yang et al., 2021).

According to family systems theory (Cox & Paley, 1997), a family consists of multiple subsystems (e.g. parent-child subsystems, parent-parent subsystems) and individual members. Additionally, there is mutual influence among these factors (Cox & Paley, 1997). Therefore, the consequences of parental burnout may spread within the family. Traditional family-centered Chinese culture places special emphasis on family harmony, and it can be anticipated that parental burnout may compromise family cohesion. However, the mediating mechanisms between parental burnout and its impact on the entire family remain unclear. Parent-child relationships constitute crucial elements within parent-child subsystems and may transmit the relationship between individual factors of parents and the entire family. Thus, this study aims to examine the mediating roles of parent-child relationships in the association between parental burnout and family cohesion.

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Parental burnout and family cohesion

From the perspective of family systems theory, the significance of the family transcends individual traits or interactions among family members (Cox & Paley, 1997; Mitchell et al., 2016). Previous research has indicated that parents who experience burnout report lower levels of family function (Yuan et al., 2022). However, the consequences of parental burnout on family cohesion have been insufficiently explored. Within the context of China's family-oriented culture, parents are often selflessly devoted to their children and prioritize maintaining family cohesion. Consequently, the threat of a breakdown in family harmony inevitably impacts the entire family, damaging family cohesion. Parents experiencing burnout may become emotionally distant from their children in an effort to alleviate emotional exhaustion (Mikolajczak et al., 2020). This emotional withdrawal disrupts connections among family members, weakening the bonds that hold the family together. Previous studies have found that when children are exposed to parents who struggle with emotional regulation, they face greater difficulty in perceiving high levels of family cohesion and adaptability (Esposito-smythers et al., 2006; Li et al., 2018). Research in the field of parenting stress has also shown that higher levels of parenting stress are linked to lower family cohesion (Mitchell et al., 2016). Given that parental burnout results from the long-term accumulation of parenting stress (Roskam et al., 2017), it is inferred that parental burnout is negatively related to family cohesion.

Parent-child relationships as mediators

Deliberating on potential mediating effects enables a better understanding of the transmission mechanisms underlying the consequences of parental burnout, thereby enhancing the feasibility of intervention and prevention strategies. Family systems theory offers valuable insights into the potential mediating roles of parent-child relationships. According to the spillover hypothesis of family systems theory (White, 1999), emotions or behaviors experienced in one subsystem can transfer to another subsystem related to the individual. In other words, parental burnout can spill over into parent-child subsystems. For example, parents experiencing high levels of parenting stress often perceive themselves as having insufficient parenting ability and, as a result, are less engaged in parent-child interactions (Crnic & Low, 2002). Notably, parent-child interaction is a primary means of maintaining a positive parent-child relationship. Parents experiencing burnout not only reduce their participation in the interactive process but also tend to withdraw from their parental roles, thereby posing a significant challenge to parent-child relationship. Previous research has shown that

parental burnout spills over into parent-child relationship (Gillis & Roskam, 2019). Both cross-sectional and longitudinal studies in the field of parenting stress have further substantiated this spillover process (Besemer & Dennison, 2018; Ho et al., 2022). The relationships between family members are the cornerstone of family harmony and stability (Minuchin, 1985). Healthy parent-child relationships enable parents and children to collaboratively shape meaningful lives through close unity and cooperation, which is crucial in consolidating family cohesion. Conversely, unhealthy parent-child relationships compromise family cohesion. Studies conducted in mainland China (Li et al., 2018) and Hong Kong (Shek, 2001) suggest that positive parent-child relationships predict higher levels of family functioning. Furthermore, family cohesion is significantly enhanced when parents improve their ability to adjust parent-child relationships (Cornett & Bratton, 2014). Overall, parental burnout is expected to influence family cohesion through parent-child relationships.

Furthermore, parent-child subsystems can be further divided into father-child subsystem and mother-child subsystem, facilitating the exploration of the crossover effects of parental burnout on parent-child relationships between partners. The crossover effect refers to the phenomenon in which emotions or behaviors experienced by one individual in a family subsystem influence the emotions or behaviors of their partner in another subsystem (White, 1999). For instance, if one parent experiences burnout, they may express dissatisfaction to their spouse regarding the challenges encountered in parenting. This can potentially affect marital quality, which, in turn, influences the relationship between both spouses and their children. Research has shown that parenting stress experienced by either the husband or wife is associated with the marital satisfaction of their partner (Deschênes et al., 2023). Among couples with lower marital satisfaction, it becomes increasingly difficult to maintain positive relationships with their children, a perspective supported by extensive research (Erel & Burman, 1995; Harrist & Ainslie, 1998; Kouros et al., 2014; Shek, 1998). A study utilizing the actor-partner interdependence model (APIM) revealed that fathers and mothers were more influenced by their partner's depressive symptoms regarding their perceptions of parental communication with their children than by their own depressive symptoms (Ponnet et al., 2013). Additionally, Wang et al. (2022) supported the crossover process, finding that father-adolescent communication mediated the relationship between mothers' phubbing and adolescents' depressive symptoms, while mother-adolescent communication mediated the relationship between fathers' phubbing and adolescents' depressive symptoms. Based on these findings, it can be inferred that

parental burnout may also impact family cohesion through the relationships with both partners and children.

Parental differences in the relationship between parental burnout, parent-child relationships and family cohesion

Exploring parental gender differences is crucial for understanding parenting roles. Roskam and Mikolajczak (2020) found that mothers generally experience higher levels of parental burnout than fathers. Similarly, Chinese mothers exhibited higher levels of parental burnout compared to fathers (Cheng et al., 2020; Ping et al., 2022). These parental differences can be attributed to the distinct divisions of labor in parenting, which align with traditional gender role expectations as described in Western studies (Wille, 1995). A similar concept exists in Chinese culture, encapsulated by the saying “nan zhu wai, nü zhu nei” (the man goes out to work while the woman takes care of the family). Parental reports and observations have also indicated that mothers are more involved in parenting, while fathers tend to work longer hours than mothers (Wille, 1995). Recent studies have noted that the rate of working mothers has increased (Wang et al., 2013). Furthermore, the father’s role in parenting extends beyond providing financial support (Jeynes, 2016). However, Latshaw and Hale (2016) suggested that mothers still spend more time on parenting after work, and parental gender differences in parenting responsibilities persist. Mothers are more engaged in their children’s daily lives, and this long-term involvement in parenting roles increases the risk of parental burnout for mothers.

There may also be inconsistencies in the effects of parental burnout on parent-child relationships and family cohesion. Previous studies have shown that mothers tend to reduce their parenting involvement, which has a greater impact on children (Bass et al., 2009; Costigan et al., 2003). As fathers gradually take on more family responsibilities (Li, 2020), we hypothesize that fathers’ parental burnout could also influence family cohesion through parent-child relationships (both father-child relationship and mother-child relationship), though the effect is expected to be weaker than that for mothers.

The present study

Previous research on the family consequences and mechanisms of parental burnout has been relatively limited, particularly in exploring parental differences. Grounded in family systems theory, the present study utilized triadic data from fathers, mothers, and children to examine the relationships between fathers’ and mothers’ parental burnout, father-child relationship and mother-child relationship, and family cohesion

(See Fig. 1). We hypothesized that fathers’ parental burnout would be negatively correlated with family cohesion (H1a), and that mothers’ parental burnout would similarly be negatively correlated with family cohesion (H1b). Furthermore, we proposed that father-child relationship and mother-child relationship would mediate the association between fathers’ parental burnout and family cohesion (H2a). Likewise, we hypothesized that mother-child relationship and father-child relationship would mediate the association between mothers’ parental burnout and family cohesion (H2b). Finally, based on social phenomena and prior literature, we investigated potential parental differences within the proposed model.

Methods

Participants

The study employed a cluster sampling method and was conducted in June 2021 across 13 junior high schools in various districts and counties in China. Well-trained psychology graduate students adhered to procedural manuals to ensure a standardized data collection process. Prior to participation, both children and their parents were informed of their right to decline participation or skip any questionnaire items they preferred not to answer. Children completed anonymous survey questionnaires at school, each assigned a unique identification number. They were instructed to record these numbers and inform their parents upon returning home. Class teachers distributed the corresponding online survey links to the parents. Parents then completed the online surveys and registered the matching identification number. Triadic data from fathers, mothers, and children were linked using these unique numbers.

Data collection involved 4,798 families, each comprising a first-year junior high school student and their parents. It should be noted that the research program measured variables beyond those used in this specific study. In the present study, fathers and mothers separately reported their levels of parental burnout, while children reported on father-child relationship, mother-child relationship, and overall family cohesion. In the process of matching father-mother-child triadic data, 2,376 families were excluded from the study because one or more family members did not complete the required questionnaires. Additionally, stepfamilies were excluded due to potential confounding factors. Research indicates that stepchildren often resist stepparents’ disciplinary efforts and exercise of authority (Nozawa, 2015). This dynamic may compel stepparents to invest more parenting resources, leading to an imbalance between resources and demands, which could induce parental burnout. To maintain consistency in understanding family processes, households

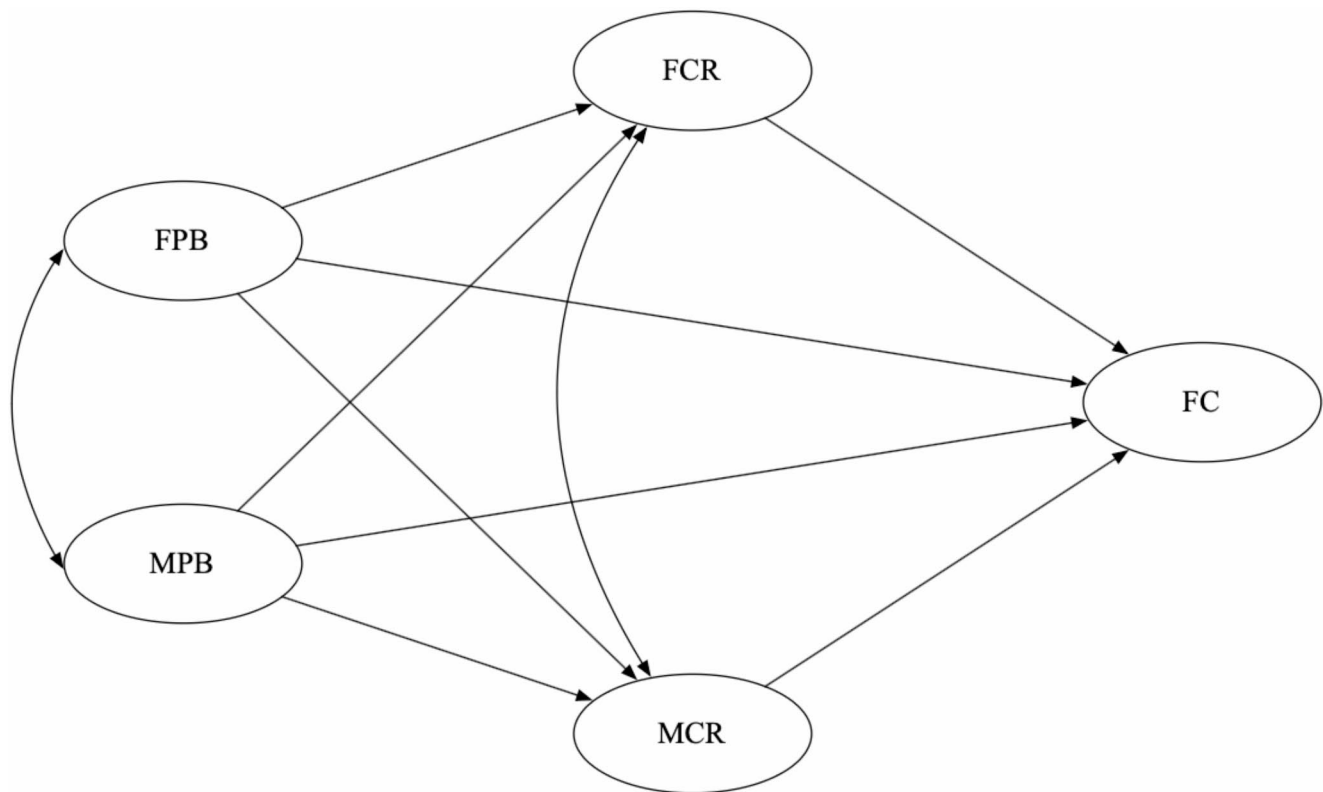


Fig. 1 Hypothesized associations between parental burnout, parent-child relationships and family cohesion. *Note.* FPB: fathers' parental burnout, MPB: mothers' parental burnout, FCR: father-child relationship, MCR: mother-child relationship, FC: family cohesion. The

primary research variables are in the form of latent variables, represented by ovals. The control variables, gender and SES, in the form of observed variables, are not presented for the sake of simplicity

where children did not reside with their biological parents were also excluded. Specifically, 37 households classified as stepfamilies and 186 families where children did not cohabit with their biological parents were removed from the analysis.

A detailed assessment was then conducted on the remaining 2,199 families to ensure data quality. A rigorous data-cleaning process was applied, including checks for "lie detection" questions and regular response patterns. Responses from 266 families were excluded because one or more family members failed to pass the quality control checks. Subsequently, data from 20 families were excluded due to more than 30% of the responses from a single family member being incomplete. Ultimately, 1,913 families were retained for the final analysis, resulting in a data efficiency rate of 87.00%.

Measures

Parental burnout

Utilizing the Parental Burnout Assessment (PBA) scale, as devised by Roskam et al. (2018), for the evaluation of

parental burnout levels. Cheng et al. (2020) translated it into Mandarin, demonstrating robust reliability and validity in testing with parents of Chinese second-year junior high school students as subjects. The PBA contained 23 items constituting four subscales: exhaustion in one's parental role, contrast with previous parental self, feelings of being fed up with one's parental role, and emotional distancing from one's children. The items were rated using a 7-point Likert scale, ranging from 1 ("Never") to 7 ("Every day"). Mean scores were calculated across all items, with higher scores indicating the greater presence of parental burnout symptoms. In the present study, Cronbach's α coefficients were 0.976 for both fathers and mothers.

Parent-child relationships

The Child-Parent Relationships Scale (CPRS), as revised by the "National Children's Study of China, NCSC" (Dong & Lin, 2011), was employed to assess the relationship between children and their parents. Children were required to report on both father-child relationship and mother-child relationship. This measurement contained eight dimensions, six of which described the satisfaction and support that

children receive from their parents, including satisfaction, intimacy, affection, instrumental help, value recognized, and companion. Two of these dimensions, annoyance and conflict, described the conflict between the children with their parents and the items from these two dimensions were reverse-scored. All items were measured using a 5-point Likert scale. Responses for items in the satisfaction dimension ranged from 1 (“Very dissatisfied”) to 5 (“Very satisfied”), while the other 22 items queried the frequency of specific events, with responses ranging from 1 (“Never”) to 5 (“Always”). Higher scores denoted a more favorable relationship between children and parents. In this study, Cronbach’s α coefficients were 0.937 for father-child relationship and 0.934 for mother-child relationship.

Family cohesion

Family cohesion was assessed using the cohesion subscale of the Family Function Scale (Bloom, 1985). The family cohesion subscale comprised 5 items, including two items scored in reverse. Children were queried about the interactions among their family members over the past year. The response options range from 1 (“Never true for my family”) to 4 (“Very true for my family”). Higher scores indicate more family cohesion children felt. This scale has demonstrated high internal consistency reliability and has been validated in studies conducted in China and other countries (Elam et al., 2018; Fosco et al., 2022; Li et al., 2009; Yamamoto et al., 2016). In the present study, Cronbach’s α for the Family Cohesion Scale was 0.753.

Covariant

The analysis included children’s gender and perceived subjective social and economic status (SES) as control variables. Subjective SES was assessed using the MacArthur Scale, a 10-rung ladder (Adler et al., 2000). The ladder represents increasing levels of education, social status, and life quality. Children were asked to select the rung that best represented their family’s situation. Higher scores on the ladder indicate higher perceived family social status.

Data analysis

A preliminary analysis was conducted using the SPSS 26.0 statistical package before testing the research hypotheses. Multicollinearity among variables was assessed using the variance inflation factor (VIF) to prevent inaccurate estimates. Descriptive statistics, including means, standard deviations, and Pearson correlation coefficients, were computed after confirming the VIF tests met the required criteria.

The structural equation model was constructed using Mplus 8.3. Missing data were coded as 999, and missing data syntax was incorporated in Mplus. All missing data were automatically estimated using the Full Information Maximum Likelihood (FIML) method, which helps avoid information loss and offers distinct advantages in enhancing reliability (Schafer & Graham, 2002). A bootstrap procedure was executed 5000 times to estimate 95% bias-corrected bootstrap confidence intervals for all hypothesized indirect effects. The presence of an indirect effect was determined by examining whether the 95% bias-corrected bootstrap confidence intervals excluded zero, indicating a significant difference between the indirect effect and zero (Zucchini, 2000). Within the SEM framework, the evaluation of model fit reflects the underlying theory (Hooper et al., 2008). In this study, model fit was assessed using the following indices: chi-square test (χ^2), relative chi-square value (χ^2/df), confirmatory fit index (CFI), Tucker-Lewis Index (TLI), root mean square error of approximation (RMSEA), and standardized residual mean root (SRMR). Given the mathematical properties of χ^2 , which tends to inflate with large sample sizes, it can lead to significant results too easily (Bentler & Bonett, 1980). Therefore, the relative chi-square value, which is corrected for degrees of freedom (df), was used to assess model fit. Following the recommendation by Wheaton et al. (1977), the relative chi-square value should be less than 5. The critical values for the other indices were as follows: CFI and TLI should be greater than 0.90 (Marsh et al., 2005), RMSEA should be below 0.08 (Schermerle-Engel et al., 2003), and SRMR should be below 0.10 (Hu & Bentler, 1995).

Results

Preliminary analysis

The diagnosis of multicollinearity revealed that the VIF among variables ranged from 1.517 to 1.840, which is below the established threshold of 5 (O’Brien, 2007). Consequently, no significant multicollinearity was observed among the variables (Dormann et al., 2013).

The correlation analysis revealed a significant negative association between family cohesion and both fathers’ parental burnout and mothers’ parental burnout. Moreover, there was a significant positive correlation between family cohesion and both father-child relationship and mother-child relationship. Fathers’ parental burnout exhibited significant negative correlations with father-child relationship and mother-child relationship. Mothers’ parental burnout also exhibited significant negative correlations with father-child relationship and mother-child relationship. In terms of

Table 1 Demographic characteristics of the sample

Children's age	Fathers' age	Fathers' level of education	Mothers' age	Mothers' level of education
13.27±0.34 (Boys:52.49%)	42.78±5.30	6.37±3.05	41.45±4.53	6.05±2.98
Educational level of parents: never attended school (1); Primary school (2); Junior high school (3); Senior high school (4); Secondary vocational technical school (5); Technical secondary school (6); Part-time junior college (7); Full-time junior college (8); Part-time undergraduate (9); Full-time undergraduate (10); Graduate and above (11)				

control variables, SES was found to have significant negative correlations with fathers' parental burnout and mothers' parental burnout. Additionally, SES exhibited significant positive correlations with father-child relationship, mother-child relationship, and family cohesion. Additionally, a notable negative correlation was observed between children's gender and father-child relationship. These correlations provided support for the proposed mediation model to some extent (Refer to Table 1 for details).

SEM analysis

According to $\chi^2(405)=2995.753$, $\chi^2/df=7.397$, CFI=0.945, TLI=0.937, RMSEA=0.060, and SRMR=0.072, the model proved fit well. For more details about factor loadings of all latent variables, see Table 2. In this study, fathers' parental burnout had a significant spillover effect on father-child relationship ($\beta = -0.078$, $SE=0.029$, $p=0.007$). The same spillover effect was also found in the association between mothers' parental burnout and mother-child relationship ($\beta = -0.215$, $SE=0.029$, $p=0.000$). No evidence supported the crossover effect of fathers' parental burnout on mother-child relationship ($\beta = -0.003$, $SE=0.029$, $p=0.917$). On the contrary, mothers' parental burnout negatively impacted father-child relationship ($\beta = -0.126$, $SE=0.029$, $p=0.000$) Fig. 2. Moreover, both father-child relationship and mother-child relationship can positively affect family cohesion ($\beta=0.360$, $SE=0.034$, $p=0.000$; $\beta=0.480$, $SE=0.035$, $p=0.000$). The direct effect of fathers' and mothers' parental burnout on

family cohesion were not significant ($\beta = -0.001$, $SE=0.022$, $p=0.967$; $\beta = -0.009$, $SE=0.022$, $p=0.700$) Table 3.

Subsequently, we conducted mediation analyses. The total effect results showed that fathers' parental burnout had no significant effect on family cohesion ($\beta = -0.031$, 95% CI = $[-0.095, 0.032]$), while mothers' parental burnout had a significant effect on family cohesion ($\beta = -0.157$, 95% CI = $[-0.229, -0.092]$). The chi-square difference test indicated that the total effect of mothers' parental burnout on family cohesion was significantly greater than that of fathers ($\Delta\chi^2(1)=5.652$, $p=0.017$). The indirect effect results indicated that fathers' parental burnout influenced family cohesion through father-child relationship ($\beta = -0.028$, 95% CI = $[-0.054, -0.005]$), and mothers' parental burnout influenced family cohesion through mother-child relationship ($\beta = -0.103$, 95% CI = $[-0.141, -0.072]$). This study also found that father-child relationship played a mediating role in the relationship between mothers' parental burnout and family cohesion ($\beta = -0.045$, 95% CI = $[-0.073, -0.022]$). In contrast, mother-child relationship did not mediate the relationship between fathers' parental burnout and family cohesion ($\beta = -0.001$, 95% CI = $[-0.030, 0.027]$). The chi-square difference test also showed that the indirect effect of mothers' parental burnout on family cohesion was significantly greater than that of fathers (MPB → MCR → FC V.S. FPB → FCR → FC: $\Delta\chi^2(1)=11.962$, $p=0.001$; MPB → FCR → FC V.S. FPB → MCR → FC: $\Delta\chi^2(1)=4.522$, $p=0.034$).

In summary, H1a was not supported, H1b and H2b were supported, H2a was partially supported, and the anticipated parental differences were identified Table 4 and Table 5.

Discussion

Recognizing the limitations of previous studies and grounded in family systems theory, this study incorporated tripartite reports from fathers, mothers, and children to examine the relationships between fathers' parental burnout, mothers' parental burnout, father-child relationship, mother-child relationship, and family cohesion. The inclusion of tripartite

Table 2 Descriptive data correlations between demographic and observed variables

	1	2	3	4	5	6	7
1.gender	1						
2.SES	—	1					
3.FPB	-0.02	-0.11***	1				
4.MPB	-0.03	-0.11***	0.57***	1			
5.FCR	-0.08*	0.26***	-0.18***	-0.19***	1		
6.MCR	0.02	0.26***	-0.16***	-0.25***	0.66***	1	
7.FC	0.05	0.25***	-0.21***	-0.24***	0.65***	0.69***	1
<i>M±SD</i>	—	5.96±1.58	1.66±1.07	1.72±1.07	3.74±0.77	3.94±0.73	3.24±0.62

FPB fathers' parental burnout, MPB mothers' parental burnout, FCR father-child relationship, MCR mother-child relationship, FC family cohesion. *** $p<0.001$; ** $p<0.01$; * $p<0.05$

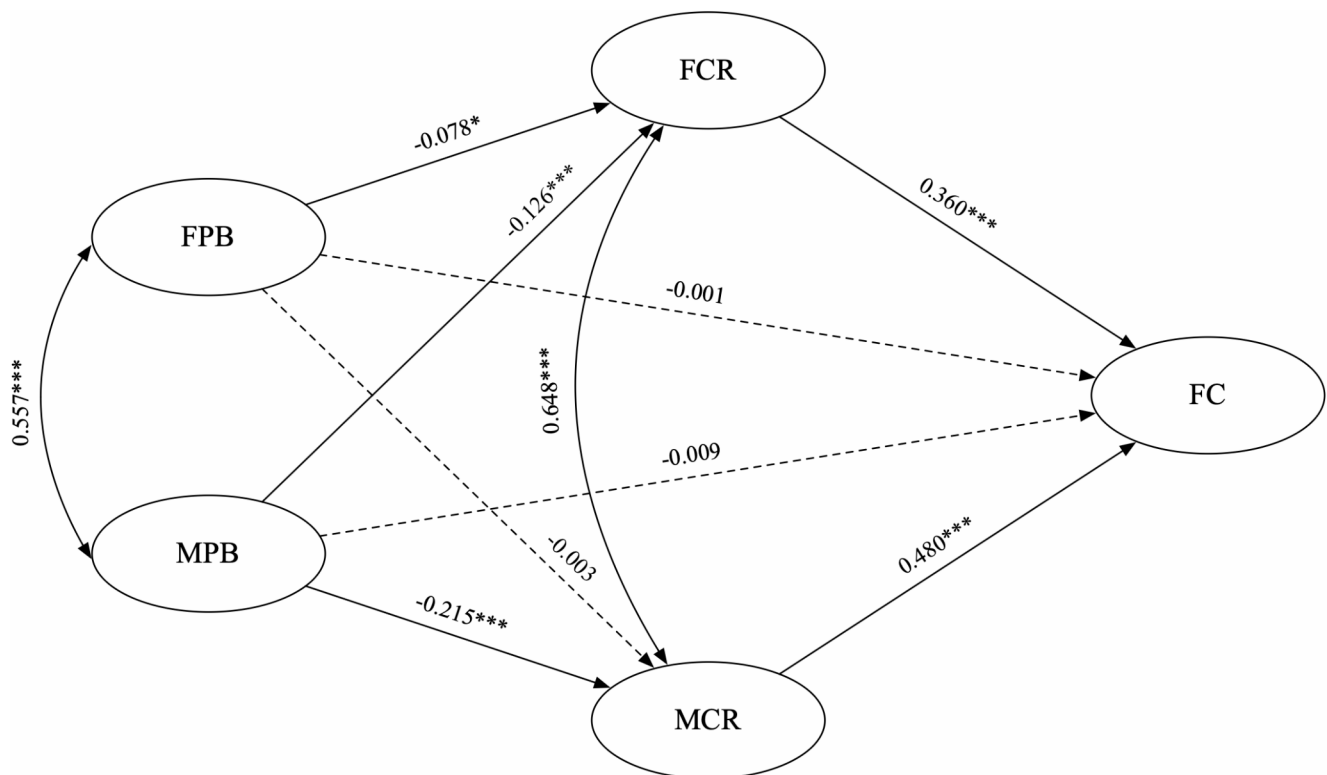


Fig. 2 Parent-child relationships as mediators of the association between parental burnout and family cohesion. *Note.* FPB: fathers' parental burnout, MPB: mothers' parental burnout, FCR: father-child relationship, MCR: mother-child relationship, FC: family cohesion.

Values on pathways represent standardized path coefficients. Dotted lines represent pathways that are not significant. The control variables are included in the analysis but not present in the figure

data enhances the validity of the results and provides a more accurate reflection of intra-family dynamics. Additionally, the study analyzed the spillover and crossover effects of parental burnout and compared parental differences to deepen the understanding of variations in parenting roles. The findings highlighted the relationship between parental burnout and family cohesion, underscoring the mediating roles of parent-child relationships, and further emphasized the severe consequences of mothers' parental burnout.

Fathers' parental burnout spilled over into father-child relationship, and mothers' parental burnout spilled over into mother-child relationship, supporting the spillover hypothesis (White, 1999). This spillover effect was also supported by Besemer and Dennison (2018), who found that mothers' parenting stress was associated with poorer subsequent parent-child relationship. The current study extends this finding by confirming that the spillover effect applies to both fathers and mothers. Subsequently, father-child relationship and mother-child relationship were linked to family cohesion. Parents who experienced parental burnout were more likely to neglect emotional support in maintaining parent-child relationships, and in extreme cases, resorted to harsh parenting style (Mikolajczak et al., 2018). These practices negatively affected parent-child relationships

(Gillis & Roskam, 2019). Negative parent-child relationships hindered the development of a warm and harmonious family environment. When children were unable to receive adequate warmth or essential support from parent-child relationships, they gradually lost confidence in the family, leading to impaired family cohesion. This finding is consistent with a recent study that indicated parent-adolescent closeness played a mediating role between paternal and maternal emotion dysregulation and family cohesion (Li et al., 2018).

Mothers' parental burnout influenced family cohesion through father-child relationship, whereas fathers' parental burnout did not spill over into mother-child relationship. Consequently, mother-child relationship did not mediate the association between fathers' parental burnout and family cohesion. These findings partly support the crossover hypothesis (White, 1999). A similar transmission pattern was observed in previous research, where mothers' hostility affected father-child relationship through crossover effects (Newland et al., 2015). The maternal gate-closing effect may help explain this phenomenon (Trinder, 2008). The gate-closing effect refers to the negative influence of mothers in hindering fathers' involvement in parenting (Trinder, 2008), which may also imply that mothers can intercept the

Table 3 Factor loadings for all latent variables

Construct	Item	Std Factor loadings	CR	AVE
Fathers' parental burnout	FPB1	0.933***	0.963	0.866
	FPB2	0.971***		
	FPB3	0.940***		
	FPB4	0.875***		
Mothers' parental burnout	MPB1	0.944***	0.957	0.847
	MPB2	0.958***		
	MPB3	0.920***		
	MPB4	0.855***		
Father-child relationship	FCR1	0.871***	0.894	0.527
	FCR2	0.475***		
	FCR3	0.675***		
	FCR4	0.782***		
	FCR5	0.381***		
	FCR6	0.860***		
	FCR7	0.759***		
	FCR8	0.838***		
Mother-child relationship	MCR1	0.863***	0.889	0.520
	MCR2	0.376***		
	MCR3	0.673***		
	MCR4	0.799***		
	MCR5	0.328***		
	MCR6	0.864***		
	MCR7	0.764***		
	MCR8	0.864***		
Family cohesion	FC1	0.893***	0.796	0.485
	FC2	0.910***		
	FC3	0.307***		
	FC4	0.798***		
	FC5	0.262***		

FPB fathers' parental burnout, MPB mothers' parental burnout, FCR father-child relationship, MCR mother-child relationship, FC family cohesion, CR composition reliability, AVE average variance extracted. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Table 4 Mediation effect analysis

Path	β	95%LLCI	95%ULCI
Total Effect			
FPB $\rightarrow$ FC	-0.031	-0.095	0.032
MPB $\rightarrow$ FC	-0.157	-0.229	-0.092
Direct Effect			
FPB $\rightarrow$ FC	-0.001	-0.050	0.047
MPB $\rightarrow$ FC	-0.009	-0.060	0.042
Indirect Effect			
FPB $\rightarrow$ FCR $\rightarrow$ FC	-0.028	-0.054	-0.005
FPB $\rightarrow$ MCR $\rightarrow$ FC	-0.001	-0.030	0.027
MPB $\rightarrow$ MCR $\rightarrow$ FC	-0.103	-0.141	-0.072
MPB $\rightarrow$ FCR $\rightarrow$ FC	-0.045	-0.073	-0.022

FPB fathers' parental burnout, MPB mothers' parental burnout, FCR father-child relationship, MCR mother-child relationship, FC family cohesion. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 5 Parental differences tests

Effect differences	$\chi^2(1)$	p
Total Effect: FPB $\rightarrow$ FC V.S. MPB $\rightarrow$ FC	5.652*	0.017
Indirect Effect: MPB $\rightarrow$ MCR $\rightarrow$ FC V.S. FPB $\rightarrow$ FCR $\rightarrow$ FC	11.962**	0.001
Indirect Effect: MPB $\rightarrow$ FCR $\rightarrow$ FC V.S. FPB $\rightarrow$ MCR $\rightarrow$ FC	4.522*	0.034

FPB fathers' parental burnout, MPB mothers' parental burnout, FCR father-child relationship, MCR mother-child relationship, FC family cohesion. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

negative parenting effects of fathers. Specifically, mothers' extensive parenting experience and dominant role in the parenting process enable them to adjust mother-child relationship more effectively, thereby mitigating the negative impact of fathers' parental burnout. Thus, mother-child relationship may be less affected by fathers' parental burnout. In contrast, fathers' parenting experience is often more limited, and as a result, father-child relationship is more susceptible to the influence of mothers' parental burnout, which in turn affects family cohesion.

The results of the parental differences tests supported traditional gender role expectations (Wille, 1995), with mothers primarily responsible for childcare while fathers played a more limited role (Doucet, 2015). Regarding the total effect, we found that mothers' parental burnout, rather than fathers' parental burnout, negatively affected family cohesion. This finding aligns with Ping et al. (2022), who observed that mothers' parental burnout significantly predicted children's depression levels, while fathers' parental burnout did not. Chi-square difference tests further revealed significant parental differences. In every daily life, mothers typically take a more active role in the parenting process, with children accustomed to the daily care provided by mothers. Mothers play a central role in the formation and maintenance of family cohesion. When mothers experience parental burnout, they may only provide basic living support to their children and become emotionally detached (Mikolajczak et al., 2019), leading children to perceive a decline in family cohesion. In contrast, fathers' limited involvement in parenting means that even if they experience burnout, children may not perceive a significant decline in overall family cohesion.

In terms of indirect effects, the mediating role of mother-child relationship in the link between mothers' parental burnout and family cohesion was significantly stronger than the mediating role of father-child relationship in the link between fathers' parental burnout and family cohesion. The increased involvement of mothers in the parenting process fosters higher levels of secure attachment between mothers and children (Williams & Kelly, 2005). However, after experiencing parental burnout, mothers' ability to maintain mother-child relationship is more likely to be compromised, leading to a significant deterioration in family cohesion.

Moreover, the maternal gate-closing effect observed in this study further suggested parental differences. Additional tests revealed that the indirect effect of mothers' parental burnout on family cohesion through father-child relationship was significantly greater than the indirect effect of fathers' parental burnout on family cohesion through mother-child relationship. This aligns with the explanation of the maternal gate-closing effect, emphasizing the central role of mothers in the parenting process (Trinder, 2008). Finally, with societal development, traditional gender roles have been increasingly challenged, and fathers are becoming more involved in parenting. As a result, paternal burnout may also begin to affect mother-child relationship.

Limitation and implications

This study has several limitations that warrant further investigation and refinement. First, due to the cross-sectional design, causal inferences cannot be drawn, and researchers should exercise caution when making directional conclusions from the results. Longitudinal studies in the future could help establish causal pathways and further investigate the parental differences identified in this study. Second, family cohesion was reported solely by children, which limits the representation of the entire family. Future research should incorporate reports on family cohesion from fathers, mothers, and children to provide a more comprehensive understanding. Third, while this study identified parental differences, further validation of these findings using latent class analysis could enhance understanding of intra-family dynamics across different family types. Finally, as the current study was based on a Chinese sample, future research should explore the spillover effects, crossover effects, and parental differences related to parental burnout in diverse cultural contexts to enhance the generalizability of the findings.

Notwithstanding these limitations, the present study not only confirmed the mediating roles of parent-child relationships in the association between parental burnout and family cohesion but also emphasized the severity of mothers' parental burnout. Based on these findings, several recommendations for parenting practices have emerged. Maintaining positive parent-child relationships can help mitigate the effects of parental burnout on family cohesion, thereby enhancing the overall well-being of the family. Given the more pronounced consequences of mothers' parental burnout, it is recommended that fathers assume greater family responsibilities. Fathers' involvement has been shown to alleviate mothers' distress (Feldman, 2007). Moreover, research has indicated that effective co-parenting can significantly reduce levels of parental burnout (Vaydich &

Cheung, 2023). Therefore, fathers should increase their support for mothers in parenting, thereby improving the availability of parenting resources. In addition to these recommendations for parenting, clinical suggestions for practitioners working with already exhausted parents are also provided. For mothers, targeted support programs could focus on addressing the unique challenges they face. These programs could include stress management training, accessible childcare services, and peer support networks specifically tailored for mothers. For fathers, practitioners could develop programs that encourage paternal involvement and enhance parenting skills. By designing interventions to meet the distinct needs of both mothers and fathers, support systems can more effectively promote family cohesion and overall well-being.

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Data availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Competing interest The authors have no relevant financial or non-financial interests to disclose.

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